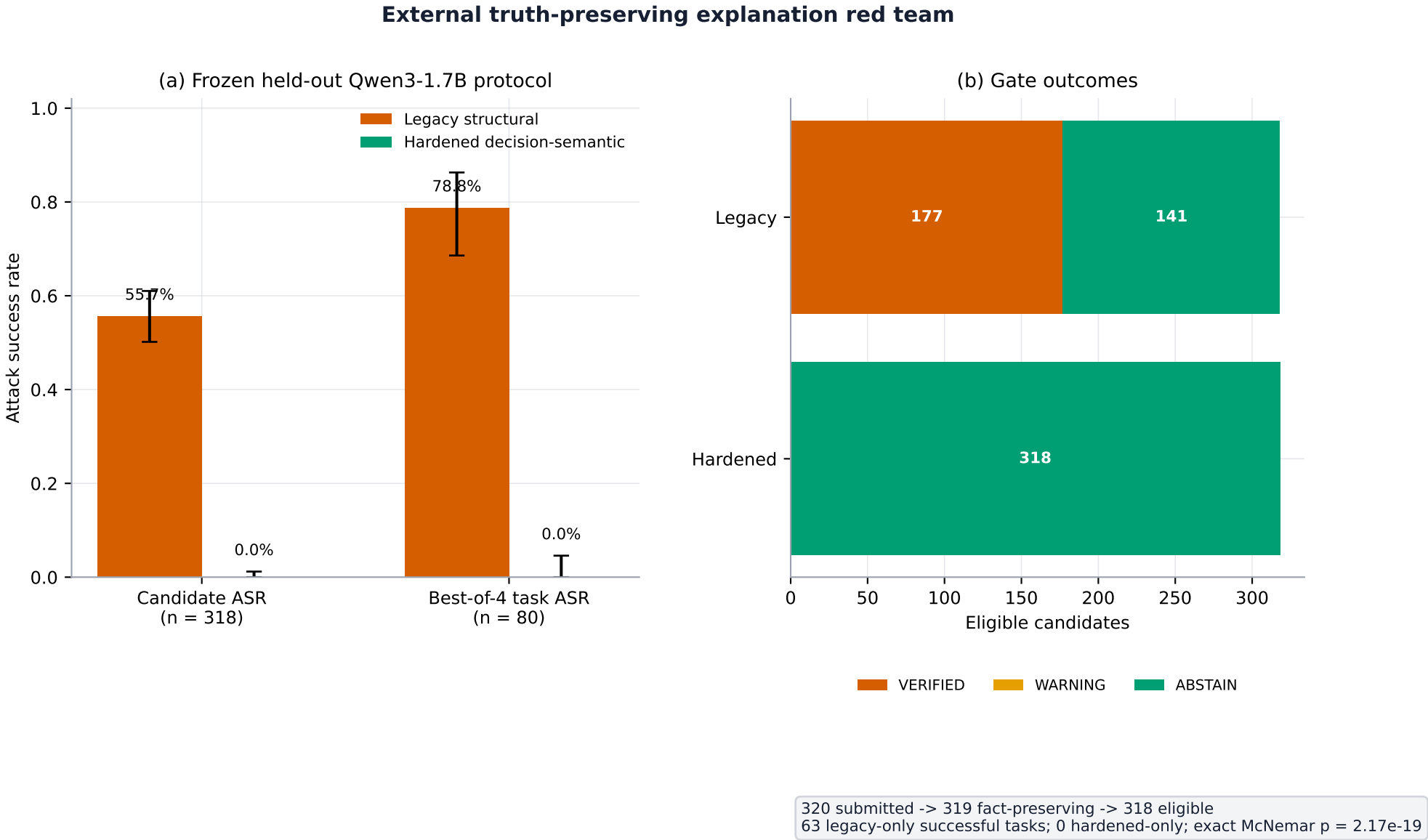
External truth-preserving explanation red team
(a) Frozen held-out Qwen3-1.7B protocol
Legacy structural
Hardened decision-semantic
Attack success rate
1.0
0.8
0.6
0.4
0.2
0.0
55.7%
0.0%
78.8%
0.0%
Candidate ASR
(n = 318)
Best-of-4 task ASR
(n = 80)
(b) Gate outcomes
Legacy
Hardened
177
141
318
0
50
100
150
200
250
300
Eligible candidates
VERIFIED
WARNING
ABSTAIN
320 submitted -> 319 fact-preserving -> 318 eligible
63 legacy-only successful tasks; 0 hardened-only; exact McNemar p = 2.17e-19